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CipherStream-128X

AXI4-Enabled High-Throughput AES Hardware Accelerator

A secure communication subsystem is being deployed in a hostile environment where all transmitted data is subject to interception and analysis. Conventional software-based encryption solutions are inadequate due to:

- High latency and limited throughput
- Susceptibility to timing-based side-channel attacks
- Inability to sustain continuous high-speed data streams

The system must process real-time streaming data including sensor feeds, telemetry, and control packets under strict constraints on Latency, Power Consumption and timing control.

Additionally, cryptographic keys must be updated dynamically without interrupting system operation. Any timing inconsistency, data corruption, or leakage is considered a critical failure.

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AXI Integration

- The cryptographic accelerator must be deployable as a standard IP block within an SoC environment, interfacing with processors, DMA engines, and memory subsystems.

To achieve this, the design must support:

- AXI4-Lite interface for configuration and control
- AXI4 Lite memory-mapped interface for high-throughput data transfer

The accelerator must ensure:

- Protocol-compliant communication under all conditions
- Sustained high-throughput operation without pipeline stalls
- Correct handling of backpressure and burst-based transfers
- Isolation between control and data paths.

Final Objective

Design a hardware-based AES-128 cryptographic accelerator IP that:

- Maintains deterministic, high-throughput encryption
- Integrates seamlessly into AXI-based SoC architectures
- Operates reliably under memory-driven workloads

Problem Statement :

Design and implement a fully pipelined AES-128 encryption engine in Verilog and integrate it as an AXI4-compliant hardware IP core capable of achieving:

- Deterministic, stall-free operation under continuous streaming conditions
- Seamless integration with AXI-based SoC systems (AXI4-Lite)
- Protocol-compliant data transfer under burst and backpressure conditions

Functional Specifications:

- AES Algorithm Requirements (AES-128 FIPS-197 Standard) :-

Block Size : 128 bits
Key Size : 128 bits
Total rounds : 10

Round Operations :-

Round 0 - AddRoundKey
Round 1-9 - SubBytes > ShiftRows > MixColumns > AddRoundKey
Round 10 - SubBytes > ShiftRows > AddRoundKey

- Key Expansion

The design must generate 11 round keys and support both Static Key Mode and Dynamic Key Update Mode.

Two architectural options are allowed:

- Precomputed Key Schedule
- On-the-Fly Key Expansion

- Modes of Operation

Mandatory:

ECB (Electronic Codebook) — for baseline verification

Optional (Bonus):

CBC (Cipher Block Chaining)

CTR (Counter Mode) (preferred for streaming systems)

Architectural Requirements :

- Fully unrolled pipeline
- Minimum 10 pipeline stages (one per AES round)
- No stalls in steady-state operation
- 32-bit wide datapath

Interface Specifications :

- The accelerator must implement standard AXI interfaces:
- Apart from AES IP core, you also have to design a slave memory which will provide the necessary input to the AES block.

NOTE: The memory should be AXI-4 lite compliant handling both the read and write requests from the AES IP core.

- Control Register: **Given through testbench**
  Start / Stop
- Key Registers: **Given through testbench**
  key[127:0] (mapped across registers)
- Status Register: **Given to testbench**
  Busy: IP core is busy, can't recieve new data.
  Idle: IP core is idle, waiting for new data.
  Error: If any error occurs.
  Done: Asserted for one cycle, after completion.

Key Interface :

- key[127:0]
- key_valid.

Key Interface Constraint :

Key updates must not corrupt in-flight data within the pipeline.

Proper synchronization between control interface and datapath is required

Interface Behavior Requirements :

- Must strictly follow AXI handshake protocol.

TESTING

Continuous high-throughput data streams

Randomized key updates

Backpressure conditions (AXI streaming behavior)

Variable-rate input/output conditions

Deliverables

- RTL Design (Verilog)
- Comprehensive Testbench, including:
  - Standard AES test vectors
  - Randomized verification
  - AXI protocol validation
- Synthesis Report:
  - LUT / FF / BRAM usage
  - Maximum frequency (Fmax)
- Performance Analysis:
  - Throughput
  - Latency
- Design Document, covering:
  - Pipeline architecture
  - S-box implementation choice
  - Key schedule strategy
  - Interface and buffering design
  - Tradeoff analysis

Evaluation Criteria

- Throughput
- Latency
- Architectural Quality
- Resource Utilization
- Must pass Standard AES test vectors (FIPS-197)
- Zero tolerance for incorrect encryption
- AXI protocol correctness

DEADLINE : 5th April

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Prize Pool:

1st Place - ₹ 5,000
2nd Place - ₹ 3,000
3rd Place - ₹ 2,000

Separate prize pool for freshers:
Goodies worth ₹ 5,000

Contact us for any queries:

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